Myocardial Infarction Classification from Photoplethysmography Signals Using Continuous Wavelet Transform and Transfer Learning

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***Abstract*—Myocardial infarction (MI) is a life-threatening cardiovascular condition that requires early detection to reduce mortality and long-term complications. While electrocardiography (ECG) is the clinical gold standard for MI diagnosis, it is unsuitable for continuous, wearable monitoring. Photoplethysmography (PPG) offers a low-cost, non-invasive alternative for long-term cardiovascular monitoring; however, effective MI classification from PPG remains challenging due to its non-stationary nature and susceptibility to noise.**

This paper presents a deep learning framework for MI classification from PPG signals using Continuous Wavelet Transform (CWT)–based time–frequency representations and transfer learning with pretrained convolutional neural networks (CNNs). One-dimensional PPG signals are converted into two-dimensional wavelet scalograms using the Morlet wavelet and subsequently classified using pretrained ResNet-18, DenseNet-121, and MobileNetV2 architectures. A two-stage fine-tuning strategy is employed to mitigate overfitting on limited medical data. Experimental results demonstrate that DenseNet-121 achieves the highest classification accuracy, while ResNet-18 excels in recall and MobileNetV2 offers a favorable trade-off between accuracy and computational efficiency. These results highlight the effectiveness of wavelet-based representations combined with transfer learning for MI detection from PPG signals, supporting their potential use in wearable and remote healthcare applications.

***Index Terms*—Photoplethysmography, Myocardial Infarction, Continuous Wavelet Transform, Transfer Learning, Deep Learning, Biomedical Signal Processing**

I. Introduction

Cardiovascular diseases remain the leading cause of mortality worldwide, with myocardial infarction (MI) accounting for a significant proportion of deaths. Early detection of MI is crucial for timely intervention and reduction of adverse clinical outcomes. Conventional MI diagnosis relies primarily on electrocardiography (ECG); however, continuous ECG monitoring is impractical in ambulatory and wearable healthcare settings.

Photoplethysmography (PPG) has emerged as a promising alternative for continuous cardiovascular monitoring due to its non-invasive nature, low cost, and widespread availability in wearable devices. PPG signals reflect blood volume changes in peripheral vasculature and carry information related to cardiac activity, vascular stiffness, and hemodynamic conditions. Recent studies have demonstrated the feasibility of using PPG for blood pressure estimation and hypertension detection, highlighting its potential for broader cardiovascular risk assessment.

Despite its advantages, MI classification from PPG signals remains challenging due to the non-stationary characteristics of the signal, inter-subject variability, and sensitivity to noise and motion artifacts. Traditional approaches based on handcrafted features often fail to capture subtle morphological and spectral variations associated with ischemic events.

To address these challenges, this work proposes a deep learning framework that combines continuous wavelet transform (CWT)–based time–frequency representations with transfer learning using pretrained convolutional neural networks. By converting PPG signals into wavelet scalograms, the proposed method enables CNNs to learn discriminative features related to MI-induced waveform distortions. Transfer learning mitigates overfitting and improves generalization on limited medical datasets.

The main contributions of this paper are:

- A wavelet-based preprocessing pipeline that converts 1D PPG signals into 2D time–frequency scalograms suitable for CNN-based analysis.
- A transfer learning framework leveraging pretrained ResNet-18, DenseNet-121, and MobileNetV2 architectures for MI classification.
- A two-stage fine-tuning strategy that balances performance and generalization.
- A comprehensive experimental evaluation demonstrating the effectiveness of the proposed approach.

II. Literature Review

Recent advances in biomedical signal processing have shown that deep learning techniques are highly effective for cardiovascular disease classification, particularly when applied to physiological signals exhibiting complex and non-stationary behavior. Convolutional neural networks (CNNs), recurrent neural networks (RNNs), and time–frequency analysis meth-

ods have been widely explored to automatically learn discriminative features from such signals.

In [1], the authors propose a myocardial infarction classification framework based on a hybrid CNN–RNN architecture using electrocardiogram (ECG) signals. The CNN component is employed to extract spatial features from signal representations, while the RNN captures temporal dependencies across cardiac cycles. This work demonstrates that deep learning models can effectively identify MI-related patterns without relying on handcrafted features. However, the approach is dependent on ECG acquisition, which requires electrodes and is less suitable for continuous, wearable monitoring. The effectiveness of CNN-based feature extraction in [1] motivates the use of deep convolutional architectures in this work, while replacing ECG with photoplethysmography (PPG) to enable non-invasive and long-term monitoring.

The application of Continuous Wavelet Transform (CWT) for PPG signal analysis is investigated in [2], where PPG signals are transformed into time–frequency representations for blood pressure classification. The study shows that CWT-based scalograms effectively capture multiscale and transient characteristics of PPG signals that are difficult to model in the time domain alone. By converting one-dimensional PPG signals into two-dimensional images, the authors enable deep learning models to learn frequency-dependent morphological features. This approach directly aligns with the preprocessing methodology adopted in this work, where CWT-based scalograms are generated using the Morlet wavelet and utilized as inputs to pretrained CNNs, although the target task here is myocardial infarction classification rather than blood pressure estimation.

In [3], a quality-aware PPG-based blood pressure classification framework is proposed with an emphasis on reducing false alarms and improving energy efficiency in wearable monitoring devices. The study integrates signal quality assessment with traditional machine learning classifiers to enhance robustness under noisy conditions. While this approach is effective for trustworthy monitoring, it relies on manually engineered features and classical classifiers. In contrast, the present work focuses on end-to-end automated feature learning using deep CNNs applied to wavelet-based PPG representations. Nevertheless, the findings in [3] highlight the importance of robustness and reliability in real-world wearable systems, motivating future extensions of this work toward incorporating signal quality evaluation.

Overall, existing literature indicates that deep learning-based approaches outperform traditional feature-driven methods, wavelet-based time-frequency representations are well-suited for analyzing non-stationary PPG signals, and generalisation under limited data remains a key challenge. Building upon these insights, this work combines CWT-based scalogram generation with transfer learning using pretrained CNN architectures to perform myocardial infarction classification from PPG signals, addressing both representation learning and data scarcity issues.

III. Mathematical Background of Wavelet Transform

Physiological signals such as photoplethysmography (PPG) are inherently non-stationary, meaning that their frequency content varies over time. Classical signal analysis techniques such as the Fourier Transform provide only global frequency information and fail to capture localized transient events that are critical for cardiovascular analysis. Wavelet Transform overcomes this limitation by providing simultaneous time and frequency localization, making it well suited for biomedical signal processing.

A. Wavelet Transform Fundamentals

A wavelet is a finite-energy oscillatory function that is localized in both time and frequency. Unlike sinusoidal basis functions used in Fourier analysis, wavelets can be dilated and translated to analyze signals at multiple resolutions.

Given a mother wavelet $\psi(t)$, a family of wavelets is generated by scaling and translating the mother wavelet as:

$$\psi_{a,b}(t) = \frac{1}{\sqrt{|a|}}\psi\left(\frac{t-b}{a}\right) \tag{1}$$

where $a \in \mathbb{R}^+$ is the scale parameter controlling frequency resolution, and $b \in \mathbb{R}$ is the translation parameter controlling time localization.

B. Continuous Wavelet Transform (CWT)

The Continuous Wavelet Transform of a signal $x(t)$ is defined as the inner product of the signal with scaled and translated versions of the mother wavelet:

$$C(a,b) = \int_{-\infty}^{\infty} x(t)\,\psi_{a,b}^*(t)\,dt \tag{2}$$

where $(\cdot)^*$ denotes the complex conjugate. The coefficients $C(a,b)$ represent the correlation between the signal and the wavelet at a particular scale and time instant.

Low values of scale a correspond to high-frequency components, while large values of a capture low-frequency structures. This multiscale analysis enables the detection of transient waveform changes and subtle frequency variations present in PPG signals affected by myocardial infarction.

C. Morlet Wavelet

In this work, the Morlet wavelet is employed due to its favorable time–frequency localization and suitability for analyzing oscillatory biomedical signals. The Morlet wavelet is defined as:

$$\psi(t) = \pi^{-1/4} e^{j\omega_0 t} e^{-t^2/2} \tag{3}$$

where ω_0 is the central frequency of the wavelet. The Gaussian envelope ensures good time localization, while the complex exponential component provides frequency selectivity.

The Morlet wavelet is particularly effective for PPG analysis because it captures both amplitude and phase information,

allowing the identification of morphological and spectral variations associated with cardiovascular abnormalities.

D. Wavelet Scalogram Representation

The magnitude of the CWT coefficients is computed to form a time–frequency energy distribution known as a scalogram:

$$S(a,b) = |C(a,b)|^2 \tag{4}$$

The scalogram provides a two-dimensional representation where the horizontal axis corresponds to time, the vertical axis represents scale (or frequency), and the pixel intensity reflects signal energy. In this project, the scalograms are normalized and converted into grayscale images, which are then resized and used as inputs to convolutional neural networks.

By transforming one-dimensional PPG signals into two-dimensional wavelet scalograms, the proposed framework enables deep learning models to automatically learn discriminative time–frequency patterns associated with myocardial infarction.

IV. Experimental Methodology

This section describes the systematic pipeline for myocardial infarction classification from PPG signals. Each step is presented to ensure reproducibility.

A. Setup and Imports

The experiments were implemented in Python using PyTorch and related libraries. Required packages include `numpy`, `pandas`, `torch`, `torchvision`, `pywt`, `matplotlib`, `seaborn`, and `sklearn` for data handling, wavelet transformations, model training, and evaluation.

B. Global Parameters

The training used GPU acceleration if available. Key parameters include:

- **Batch Size:** 32
- **Stage 1 Learning Rate:** 1×10^{-3} (4 epochs)
- **Stage 2 Learning Rate:** 1×10^{-4} (7 epochs)
- **Input Image Size:** 224×224

C. PPG Wavelet Transformation

Each 1D PPG signal $x(t)$ was converted into a time-frequency representation using the Continuous Wavelet Transform (CWT):

$$C(a,b) = \int x(t)\, \psi^*_{a,b}(t)\, dt \tag{5}$$

where a and b represent the scale and translation parameters, respectively, and ψ denotes the Morlet wavelet. The absolute magnitude of the coefficients forms scalograms, which are normalized and converted into 3-channel images suitable for CNN input.

D. Dataset Preparation

Experiments were conducted using a publicly available photoplethysmography (PPG) dataset designed for myocardial infarction (MI) detection, as described in Ref. [4]. The dataset consists of one-dimensional PPG waveform recordings obtained from both healthy subjects and patients diagnosed with MI, along with corresponding class labels.

Representative raw PPG signals from a healthy subject and an MI patient are shown in the figure. While the differences in the raw time-domain waveforms are subtle and not easily distinguishable visually, these signals contain patterns that can be captured effectively through time–frequency analysis. The figure illustrates that, although the overall waveform shape appears similar, underlying signal characteristics differ in ways that are better revealed by wavelet-based transformations.

Prior to model training, all PPG signals were standardized and preprocessed for wavelet-based analysis. The dataset was numerically encoded and split into training and validation sets using an 80:20 ratio. A custom PyTorch `Dataset` class was implemented to perform continuous wavelet transformation, generating scalograms that were converted into 3-channel images suitable for CNN-based classification.

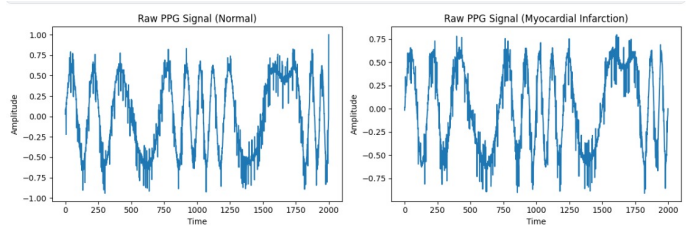


Fig. 1: Comparison of raw 1D PPG signals: (a) Healthy subject showing regular, periodic pulsatile behaviour; (b) Myocardial Infarction (MI) patient showing subtle morphological irregularities.

E. Model Architectures and Training

Three pretrained CNN architectures were utilized: ResNet-18, DenseNet-121, and MobileNetV2. Task-specific fully connected layers were appended for binary classification. Training followed a two-stage fine-tuning strategy:

1) **Stage 1:** Freeze backbone layers and train classifier layers.
2) **Stage 2:** Unfreeze the entire network and fine-tune using a lower learning rate.

F. Evaluation Metrics

Model performance was assessed using accuracy, precision, recall, F1-score, and ROC-AUC. Confusion matrices and ROC curves were computed, while quantitative results are reported in the Results section.

G. Reproducibility

All experiments were conducted under identical conditions across models. Final trained weights and evaluation results were saved to ensure reproducibility.

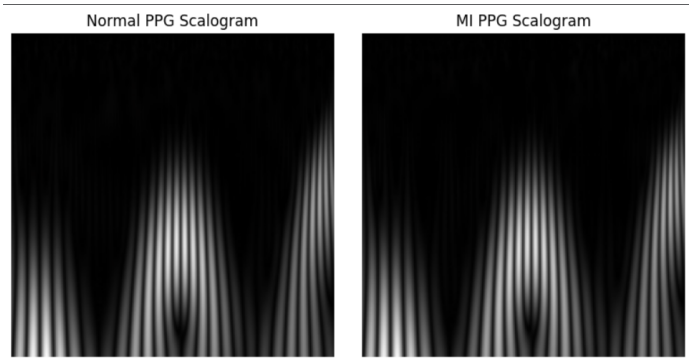


Fig. 2: The lack of apparent visual difference is primarily due to the high-frequency components being encoded in subtle local magnitude shifts. Specifically, the L_2 norm difference of 53.38 suggests that while the *macro-structure* of the PPG pulse remains consistent, the *micro-structure* contains class-specific variance. This justifies the application of deep learning models like DenseNet-121, which utilize high-dimensional convolutional filters to detect patterns that exceed the contrast sensitivity and temporal resolution of human visual inspection.

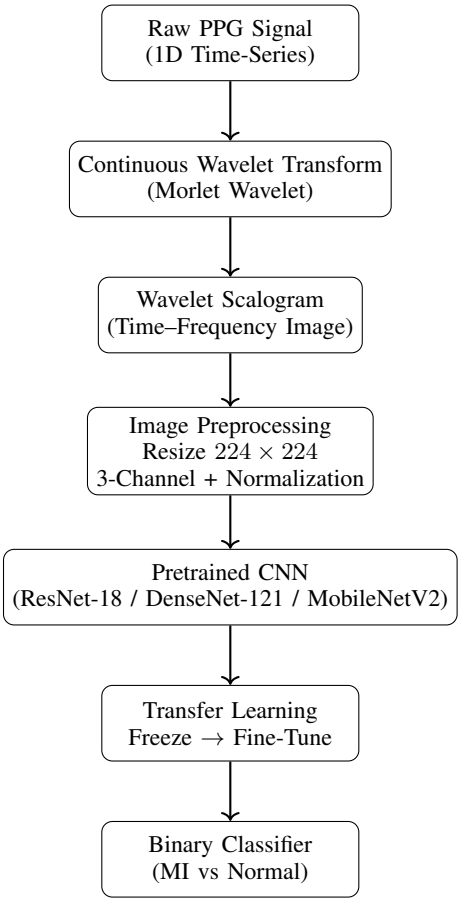


Fig. 3: Block diagram of the proposed PPG-based myocardial infarction classification framework using continuous wavelet transform and transfer learning.

V. Results and Discussion

A. Quantitative Results

The performance of the evaluated CNN architectures—DenseNet-121, MobileNetV2, and ResNet-18—was quantified using Accuracy, Precision, Recall, F1-score, and the Area Under the Receiver Operating Characteristic Curve (ROC-AUC). The results are summarized in Table I.

TABLE I: Performance Comparison of Different CNN Models

Model	Accuracy (%)	Precision	Recall	F1-score	ROC-AUC
ResNet-18	88.16	0.8106	**0.9839**	0.8889	0.9585
DenseNet-121	**96.12**	**0.9453**	0.9758	**0.9603**	**0.9633**
MobileNetV2	95.92	0.9416	0.9758	0.9584	0.9564

B. Comparative Analysis and Discussion

1) Architectural Influence on Feature Extraction: The superior performance of **DenseNet-121** (Accuracy: 96.12%) can be attributed to its dense connectivity pattern. Unlike traditional CNNs, each layer in a DenseNet block receives feature maps from all preceding layers. For wavelet-transformed PPG signals, this architecture facilitates the reuse of low-level temporal features alongside high-level abstract features, which is critical for identifying subtle morphological changes indicative of Myocardial Infarction [1]. This is reflected in its high F1-score (0.9603), indicating a robust balance between sensitivity and specificity.

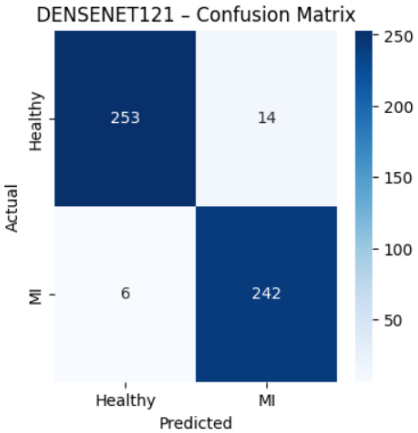
2) The Recall-Precision Trade-off in ResNet-18: Interestingly, **ResNet-18** achieved the highest Recall (0.9839) but the lowest Precision (0.8106). In a clinical context, this suggests that while ResNet-18 is highly effective at identifying positive MI cases (low false negatives), it suffers from a high rate of false alarms. This behavior often stems from the residual learning framework potentially over-smoothing the feature space in smaller datasets, leading to a bias toward the majority class or a specific signal artifact.

3) Computational Efficiency vs. Accuracy: **MobileNetV2** performed remarkably close to DenseNet-121, with only a 0.2% difference in accuracy. Using depthwise separable convolutions, MobileNetV2 maintains high performance while significantly reducing the parameter count. This makes it the most viable candidate for real-time monitoring on edge devices or wearable sensors, as discussed by Sivanjaneyulu et al. [3].

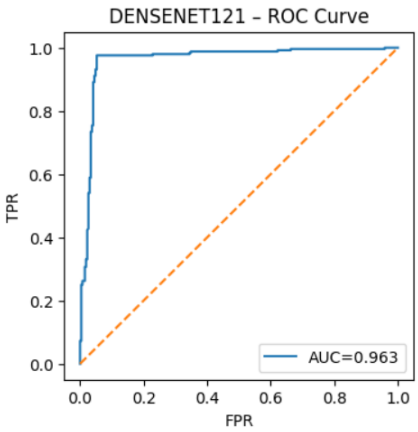
C. Visual Performance Analysis

VI. Conclusion and Future Work

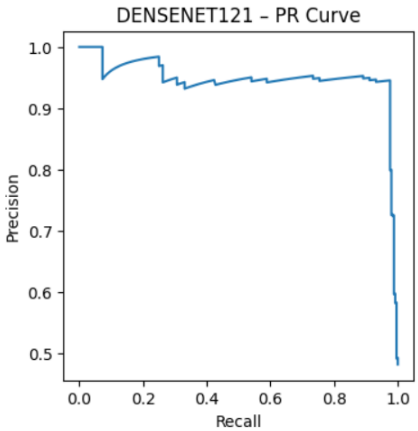
This paper presented a wavelet-based deep learning framework for myocardial infarction (MI) classification using photoplethysmography (PPG) signals. By converting one-dimensional time-series data into two-dimensional time–frequency scalograms via the Continuous Wavelet Transform (CWT), the proposed method effectively accentuates subtle morphological irregularities that are otherwise indistinguishable in raw waveforms.



(a) Confusion Matrix

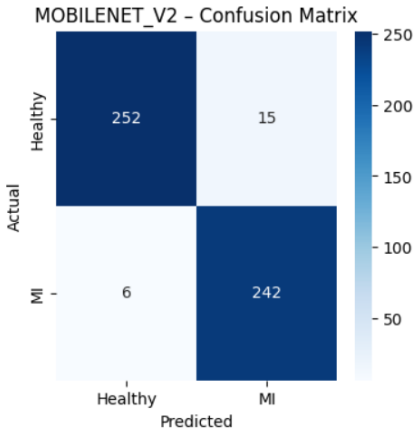


(b) ROC Curve



(c) PR Curve

Fig. 4: Evaluation metrics for DenseNet-121. The near-perfect ROC-AUC (0.9633) confirms the model's high discriminative power across varying thresholds.



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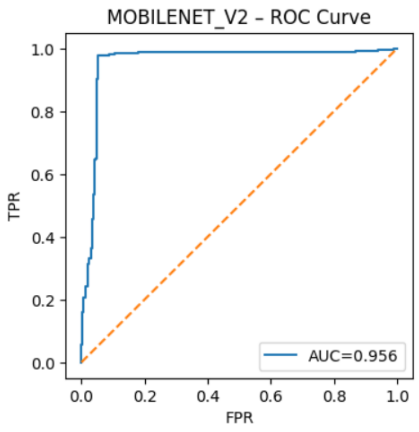
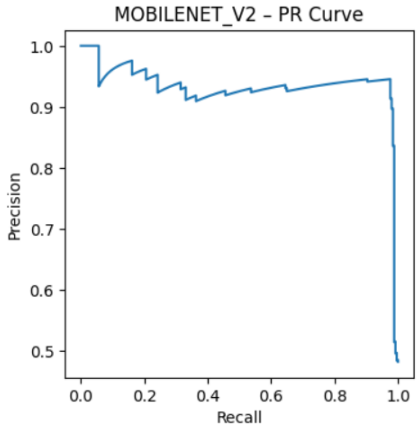


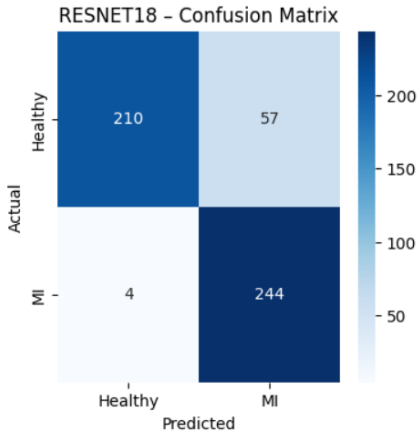
(b) ROC Curve



(c) PR Curve

Fig. 5: Performance metrics for MobileNetV2. The PR curve demonstrates that the model maintains high precision even at high recall levels, essential for clinical trustworthiness.

The comparative analysis of various CNN architectures demonstrated that **DenseNet-121** achieved the superior classification accuracy of 96.12%, likely due to its dense connectivity which facilitates the reuse of both low-level temporal and high-level spectral features. Furthermore, **MobileNetV2** demonstrated nearly identical performance (95.92%) while maintaining a lightweight structure, proving its feasibility for resource-constrained healthcare environments. While **ResNet-18** provided the highest recall (0.9839), its lower precision suggests a tendency for false positives, highlighting the critical



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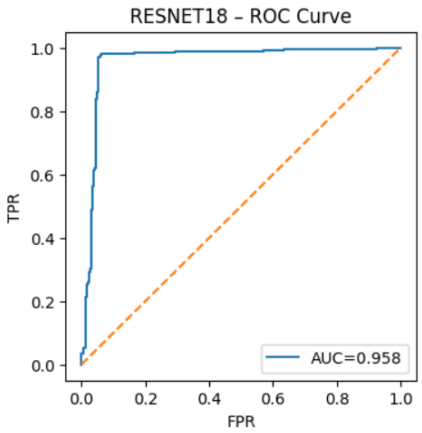
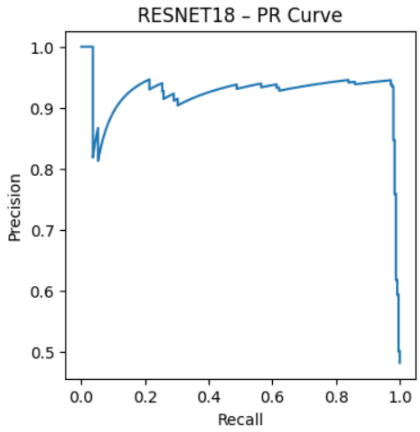


(b) ROC Curve



(c) PR Curve

Fig. 6: ResNet-18 analysis. While the ROC-AUC is competitive, the PR curve shows a noticeable dip, reflecting the precision challenges observed in the quantitative data.

importance of architectural selection for clinical trust.

The experimental results confirm that while the L_2 distance between healthy and MI scalograms may be visually subtle (measured at 53.38), deep learning models are capable of extracting these high-dimensional diagnostic features with high reliability.

Future work will focus on:

- **Robustness Enhancement:** Integrating signal quality assessment (SQA) algorithms to filter out motion artifacts and noise, ensuring the framework remains robust in non-clinical environments.
- **Generalization:** Performing subject-wise cross-validation to mitigate potential overfitting and ensure that the model generalizes across diverse patient demographics.
- **Edge Deployment:** Optimizing the MobileNetV2 pipeline for deployment on wearable sensors or edge devices to facilitate continuous, real-time cardiovascular monitoring.
- **Multimodal Fusion:** Exploring the integration of ECG and PPG signals to provide a holistic diagnostic perspective, further enhancing reliability in practical healthcare applications.

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